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Good Science

Measuring What's Right, Not What's Easy To Measure



Julian Frattini, Ph.D.
Postdoctoral Researcher
Chalmers University of Technology, Sweden
<https://julianfrattini.github.io/>



JulianFrattini/**bda4sci**

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Motivation

The Status Quo and Why We Need to Go Beyond

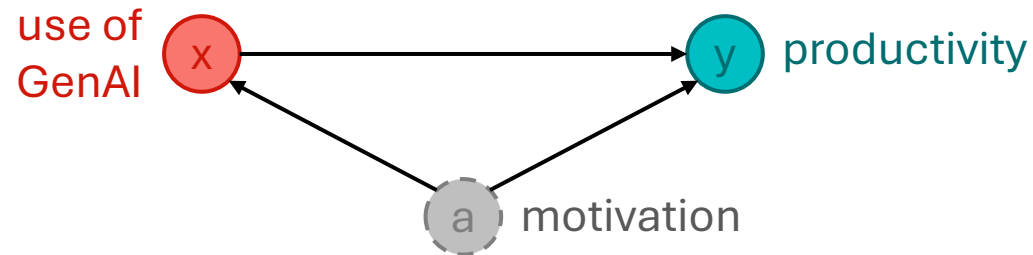
Rigorous Research

So far, we have determined that good, *rigorous* research needs to take into account two dimensions of validity:

- **Internal validity**, i.e., distinguishing actual causal effects from spurious associations, and
- **External validity**, i.e., making sure that findings actually apply not only to the studied sample, but the target population of interest.

Framing Causal Assumptions

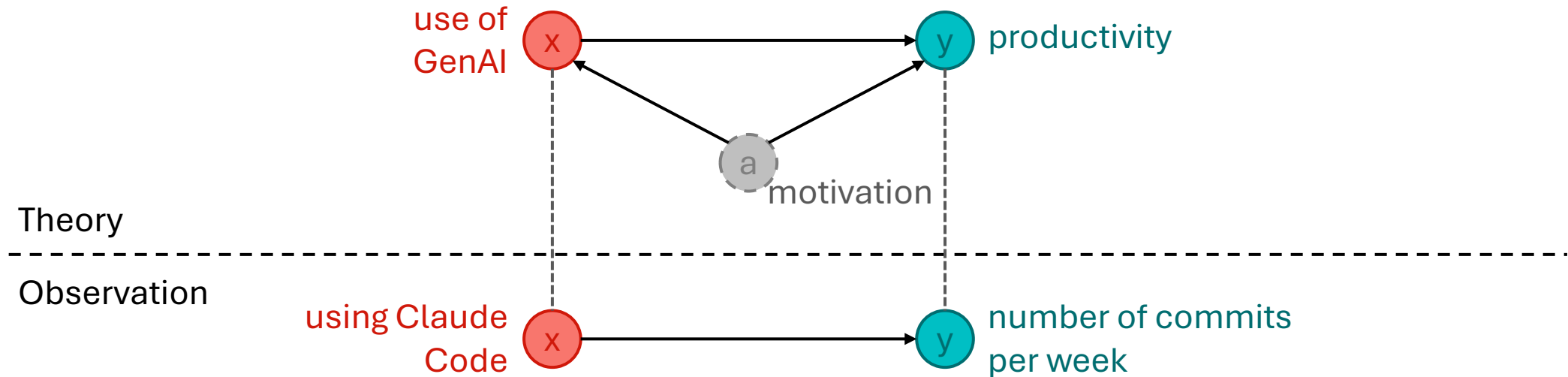
Via **causal modelling**, we make our causal assumptions explicit.



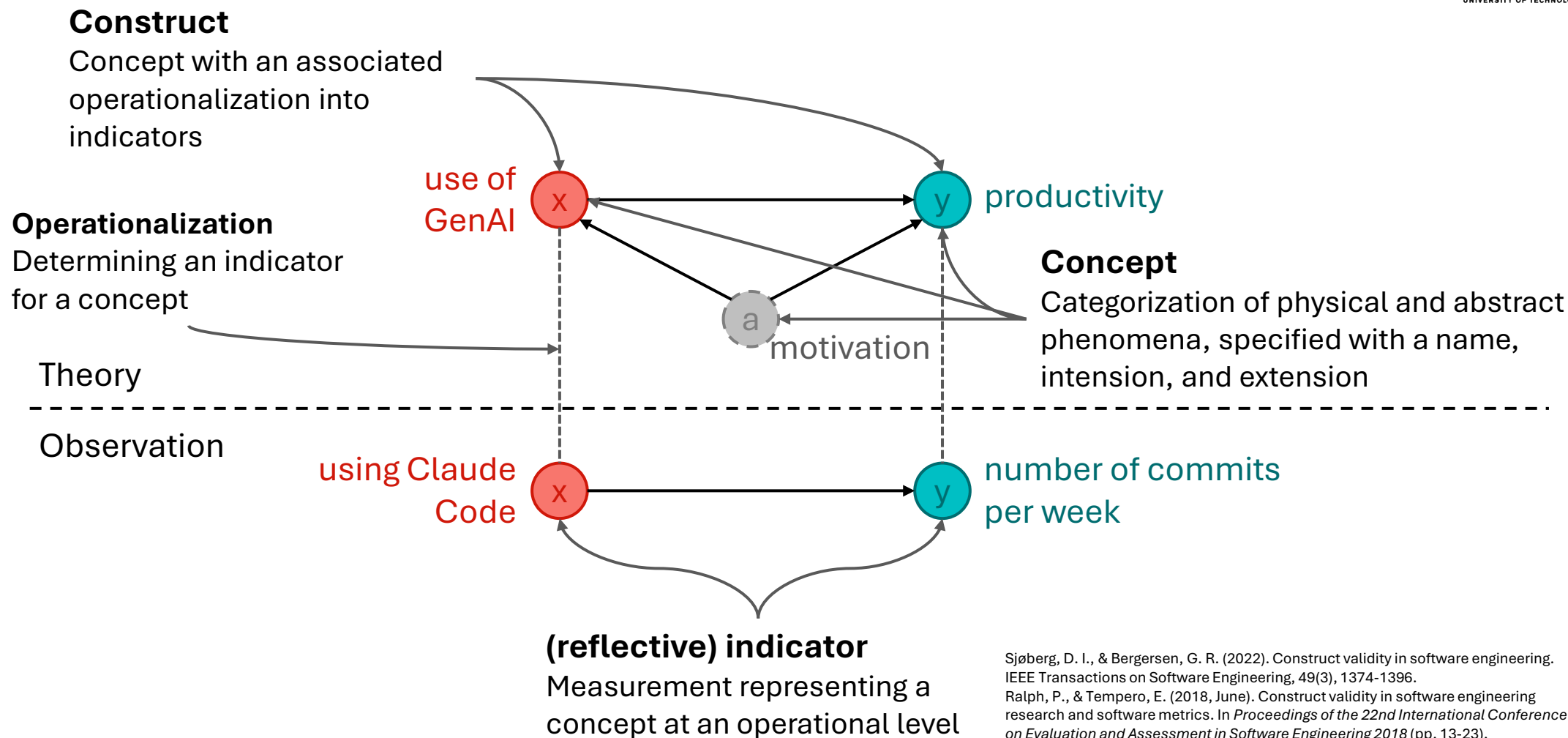
This helps communicate our causal assumptions, invites scrutiny, and helps iteratively improving them.

Levels of Research

We can distinguish two separate yet related levels:

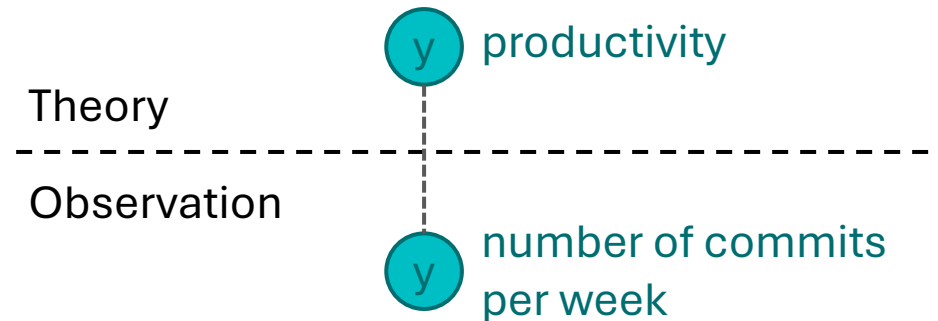


Levels of Research



Operationalization

The process of **operationalization** seems trivial, but what makes a *good* operationalization?



A common pitfall is *measuring what is easy to measure, not what is relevant to measure.*

How do you measure

Productivity?

How do you measure

Motivation?

How do you measure

Skill?

Operationalization

For every non-trivial construct, there are several options for operationalization, for example:

- **Productivity:** number of commits pushed, number of lines of code pushed, self-reported productivity, ...
- **Motivation:** self-reported motivation, number of consecutive work hours, number of tasks addressed in a day, ...
- **Skill:** self-reported skill, number of years of experience, highest academic degree, ...



Construct Validity

Definition and Assessment

Construct Validity

Next to internal and external validity, construct validity is another dimension of validity that matters to the reliability of the results.



Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236, p. P3). Berlin: Springer.

Construct validity concerns generalizing the result of the experiment to the concept or theory behind the experiment.

Threats to Construct Validity

To understand what makes a good operationalization, we can also look into what makes a *bad* operationalization via **threats to construct validity**, including

1. Inadequate Definition of Concept
2. Construct Underrepresentation
3. Construct-Representation Bias

Inadequate Definition of Concept

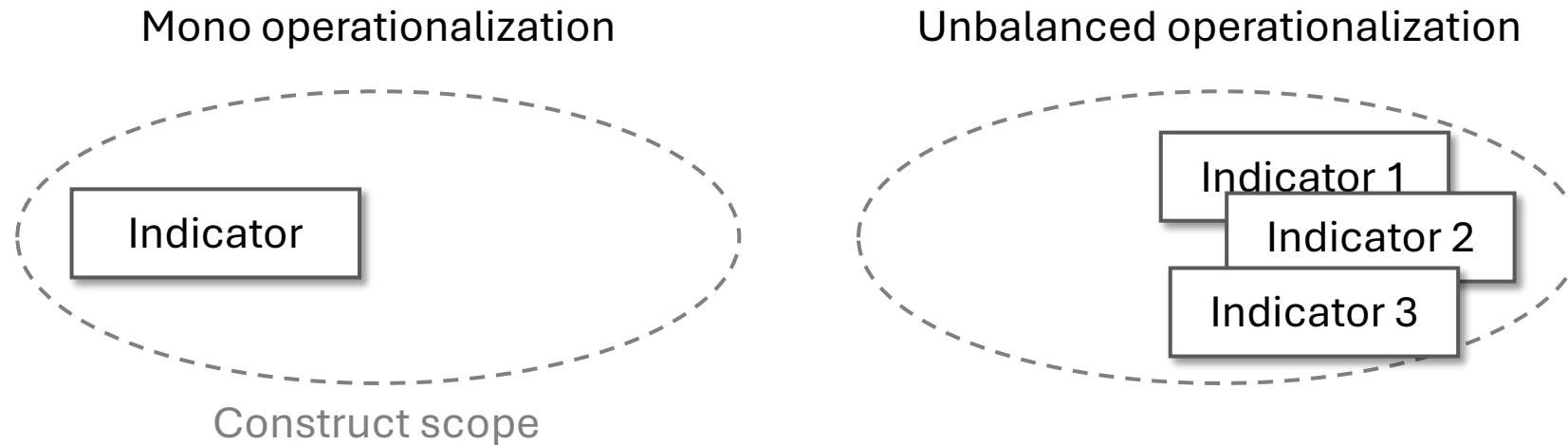
An operationalization may suffer from an *inadequate definition of a concept* if the concept is defined at too **general** a level, at too **specific** a level, or the concept is **not the right one**, even though it is at the appropriate level of abstraction.

For example, is *lead time*

- the amount of time between the proposal for a new feature or another request and its deployment in the customer's environment, or
- the amount of time that passes from the moment that the development team receives a request to the moment that it completes the work item.

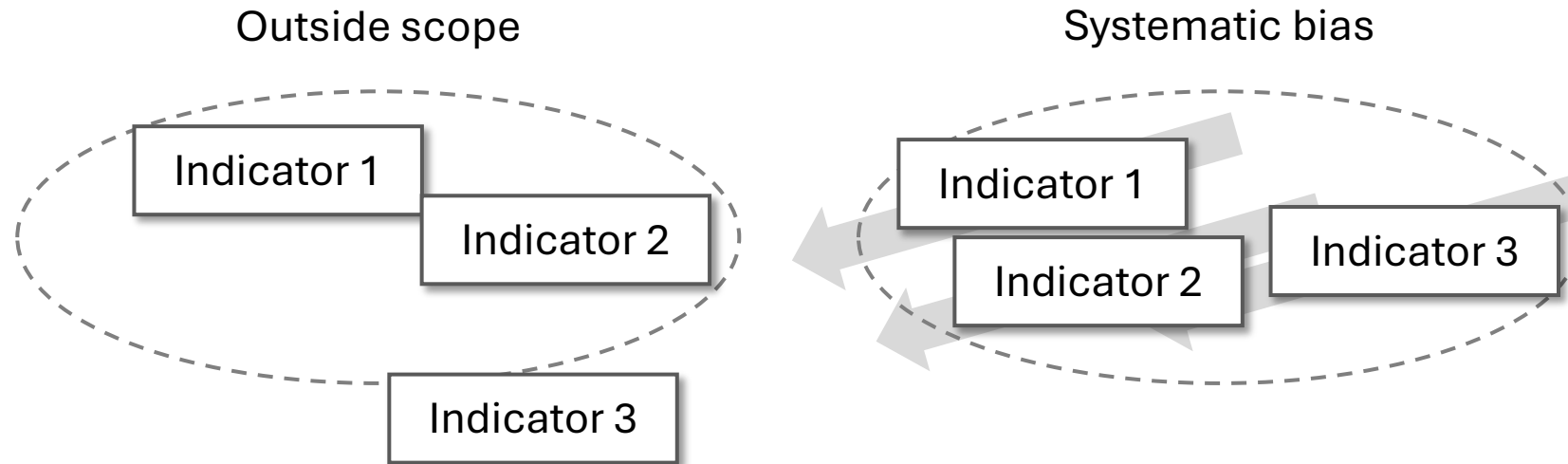
Construct Underrepresentation

Construct underrepresentation occurs when the set of indicators resulting from the operationalizations are **too narrow** to include all important aspects or dimensions of the concept that should be included.



Construct-Representation Bias

An operationalization may suffer from a *construct-representation bias* if the chosen indicators emphasize specific aspects of the construct while ignoring others.



Construct Validity

Still, none of this answers the question: *how do I reliably assess whether I am measuring the concept of interest right?*

” There is no analytical way of ensuring construct validity.

Follow best practices and be content with statements about the operational, not the theoretical level.

Construct Validity

There is no *absolute* way of measuring construct validity, but at least there is a *relative* way:

- **Convergent validity:** a measure should correlate with other (established) measures of the same construct
- **Discriminant validity:** a measure should not correlated with highly distinct constructs

Positive Examples

One of the best-known and –elaborated examples in software engineering is the operationalization of **technology acceptance** (i.e., usability and ease-of-use).

ID	Question
PU1	Using <i>system</i> in my job would enable me to accomplish tasks more quickly.
PU2	Using <i>system</i> would improve my job performance.
PU3	Using <i>system</i> in my job would increase my productivity
PU4	Using <i>system</i> would enhance my effectiveness on the job.
PU5	Using <i>system</i> would make it easier to do my job.
PU6	I would find <i>system</i> useful in my job.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS quarterly, 13(3), 319-340.



The Effects of Construct Validity

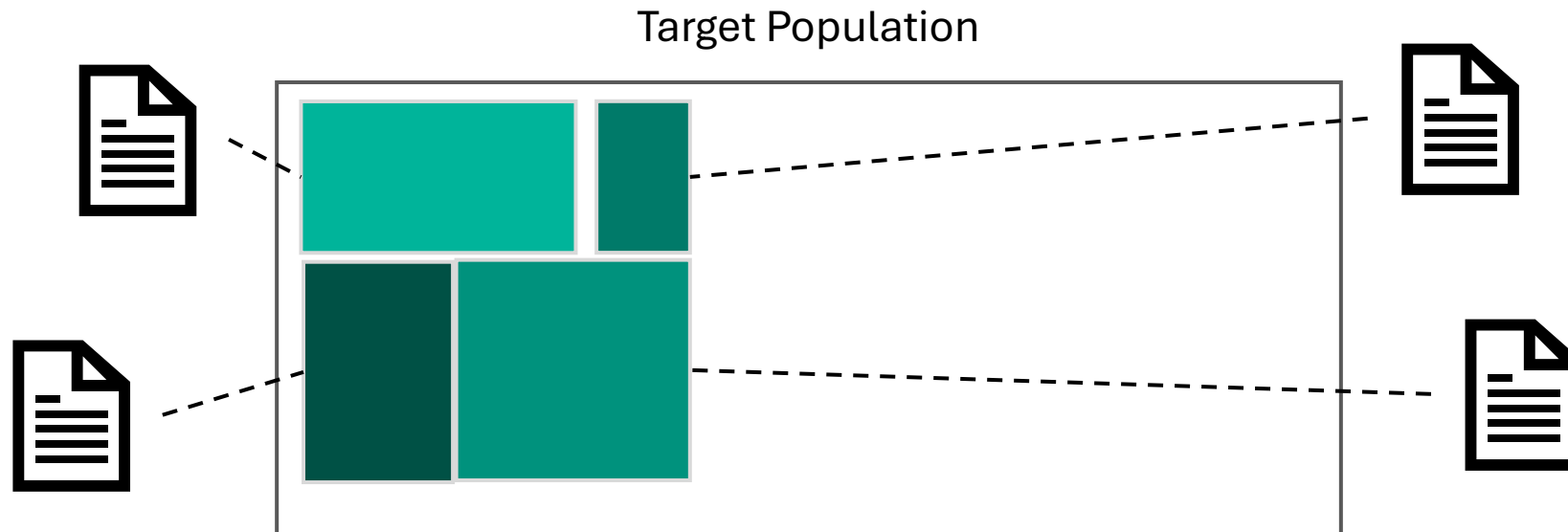
Aggregation of Studies

We have previously established that conducting good, *rigorous* research is difficult. This results in a very constrained scope of individual studies.



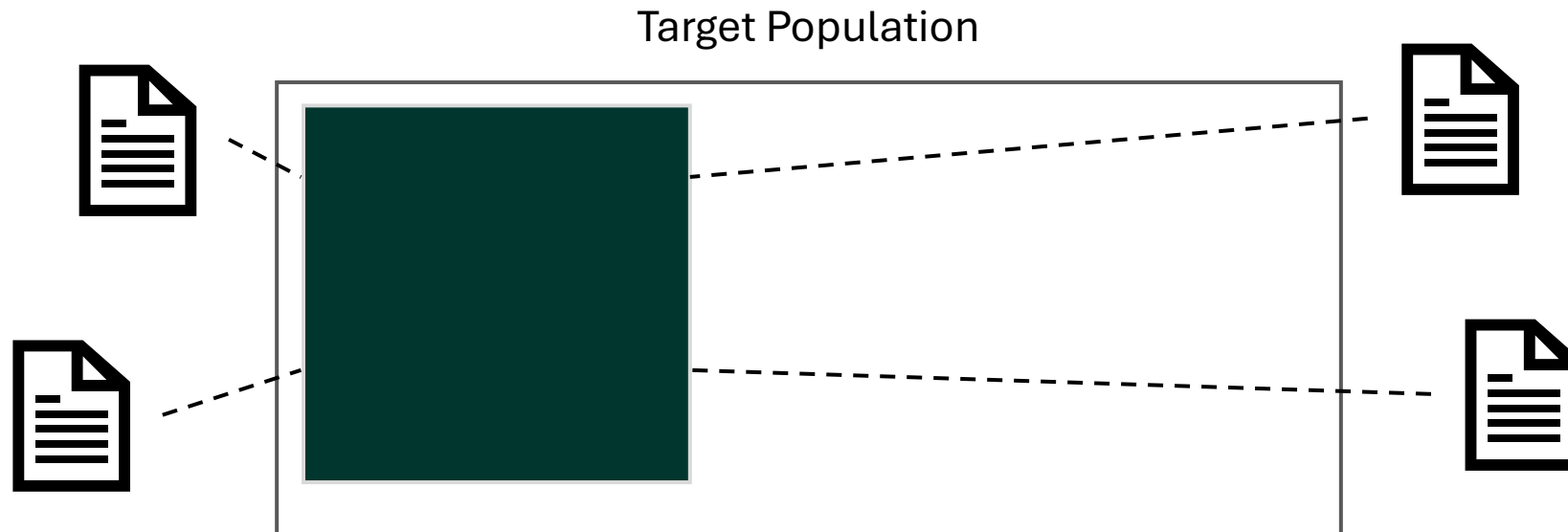
Aggregation of Studies

However, through the process of **replication**, individual studies can incrementally cover the target population in a way that would be impossible for a single study.



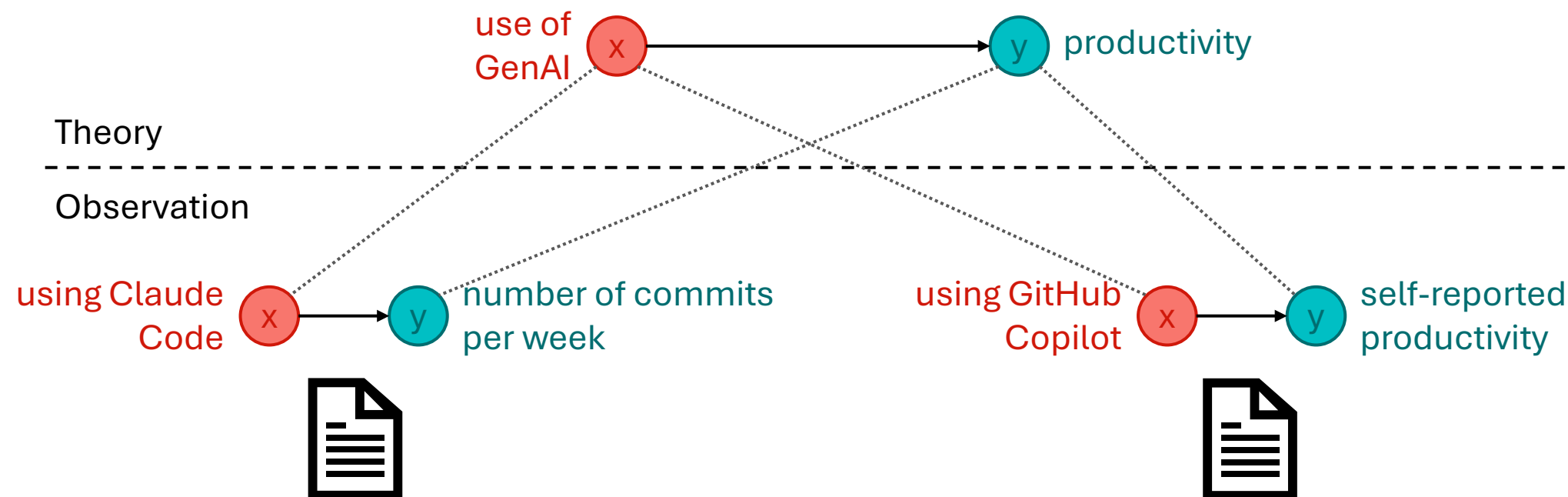
Aggregation of Studies

Several individual studies with constrained scope can be aggregated via a **meta-analysis** to one result covering a larger part of the target population.



Aggregation of Studies

For meta-analysis to work, it is essential that all individual studies operationalize all concepts in the same way.



Achieving Aggregation

Other disciplines (like medical science) heavily rely on the aggregation of individual pieces of evidence. They achieve this via

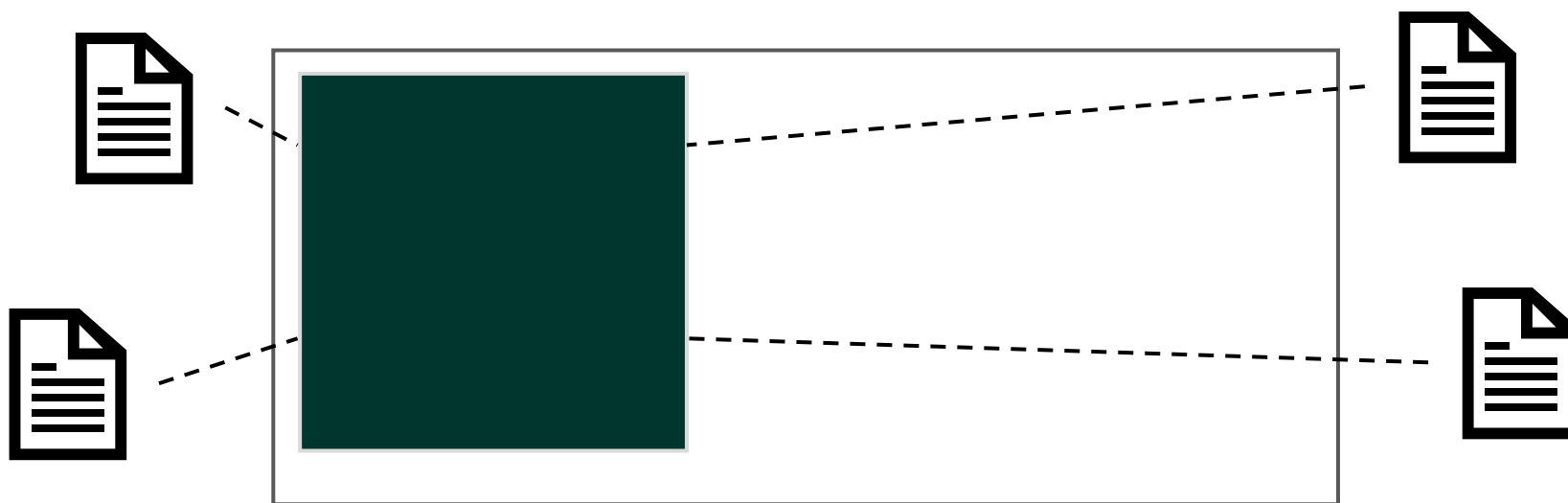


Procotols, i.e., strict guidelines on how to operationalize and measure certain variables of interest

On the other hand: stricter protocols limit creativity and exploration. Many phenomena in software engineering are not yet explored well enough to prescribe how to measure them.

Side Note on Replications

Aggregation of evidence first requires the accumulation of evidence.

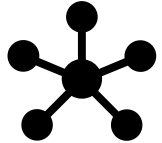


Before we can actually aggregate any evidence, somebody needs to *perform* those replication of a first study.

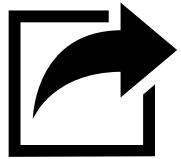


Conclusion

How to Good Science



Internal validity requires visualizing and communicating your causal assumptions, and closing any backdoor paths by controlling confounders



External validity requires awareness of potential moderators that also affect trial eligibility



Construct validity requires operationalizing constructs with an encompassing and representative set of indicators